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The Internal Structure of R-Loci

Index-Reference Alignment in Sign Languages

Abstract

Spatial reference is a central grammatical resource in sign languages and is commonly modeled by means of R-loci. While their descriptive role is well established, their grammatical status and their contribution to argument licensing remain unclear. This paper argues that R-loci should be analyzed as syntactic objects with internal structure, consisting of an index component and a referential component. On this basis, the paper proposes "Locus-Checking", a uniform syntactic operation that aligns index and reference information across R-loci in order to license verbal arguments. Within this framework, a range of phenomena traditionally treated as heterogeneous can be derived from differences in the internal configuration of R-loci, including apparent agreement effects, first-person interpretations with body-anchored verbs, and null arguments with discourse-dependent or generic readings. The proposal is grounded in empirical work on German Sign Language (DGS), which serves as the primary empirical point of reference for the development of the model, but it is intended to contribute to the general grammatical analysis of R-loci in sign languages. Rather than appealing to agreement features, null pronominal categories, or purely discourse-pragmatic licensing, the analysis derives argument realization from the internal structure of R-loci and from constrained principles governing their interaction.

1. Introduction

A central grammatical feature of sign languages is the use of spatially anchored positions, known as **R-loci**, which facilitate the tracking of discourse referents. Since their initial description, R-loci have played a central role in analyses of pronominal reference, agreement, deixis, and discourse cohesion in sign languages. Despite their empirical prominence, however, their grammatical status remains surprisingly unclear.

Across the literature, R-loci have been treated as abstract indices, agreement markers, pronominal elements, or discourse-gestural devices. Each of these perspectives captures genuine aspects of the phenomenon, yet none provides a fully explicit account of what an R-locus *is* as a grammatical object, nor of how R-loci participate in the licensing and interpretation of arguments.

As a result, a range of core phenomena in sign languages such as German Sign Language (DGS) have been treated as analytically distinct. This includes null arguments, apparent verb-class differences, and the distribution of elements such as INDEX and PAM ("person agreement marker"). Although these phenomena are empirically tightly interconnected, they are typically accounted for by invoking separate mechanisms, ranging from agreement-based analyses to null pronominal categories and discourse-pragmatic inference. As a consequence, closely related patterns are captured by formally independent mechanisms rather than by a unified structural account.

This paper argues that this heterogeneity does not directly follow from the empirical facts themselves. Instead, it reflects a missing level of structural representation. The proposal is that **R-loci themselves are syntactic objects with internal structure**, and that argument licensing in DGS can be derived from the way these objects are structurally completed. Specifically, it will be argued that every R-locus consists of two distinct but inherently connected components: an **index component**, which may be specified by a spatial coordinate, and a **referential component**, which links the locus to a discourse referent. Variation in argument realization and interpretation arises not from different licensing mechanisms, but from differences in how these two components are specified and aligned in the course of the derivation.

Throughout this paper, the term 'R-locus' is used in a structurally extended sense. Rather than referring only to spatially established discourse referents, R-loci are taken to underlie verbal argument positions in general, independently of whether they are overtly localized or referentially specified.

On this basis, the paper introduces **Locus-Checking** as a general structural operation. Locus-Checking is not a feature-checking relation in the traditional sense, nor is it merely a post-syntactic interpretive process. Rather, it is a syntactic operation that aligns partially specified R-loci in order to complete missing indexical or referential information. An argument counts as licensed if and only if the R-locus associated with the corresponding argument position becomes referentially complete through Locus-Checking. Phenomena traditionally analyzed in terms of agreement, null arguments, generic reference, or person-related effects consistently emerge as surface manifestations of a single underlying mechanism.

This approach leads to a fundamental reanalysis of several long-standing assumptions. Apparent verbal agreement is reconceived as a restricted case of Locus-Checking between R-loci that share indexical information. Null arguments are treated as structurally present argument positions whose interpretation depends on the availability and relative position of potential referents. With body-anchored verbs, first-person interpretations arise when the signer's body provides a preferred source of reference. In all cases, argument realization reduces to differences in the structural specification of R-loci and in how these specifications are aligned during Locus-Checking.

To this end, the goal of the paper is therefore not to extend existing descriptive notions of R-loci, but to **reconceptualize them as formally articulated grammatical objects** and to demonstrate that this reconceptualization yields a more unified and restrictive account of argument structure and interpretation.

The theoretical proposal developed in this paper was originally motivated by a qualitative empirical analysis of German Sign Language (DGS) conducted in the context of my MA thesis (Nazar 2025). While the present work does not present this empirical material in detail, it builds on the insights gained from that analysis and integrates them with findings and observations reported in the broader sign language literature.

The paper is structured as follows. Section 2 reviews descriptive and theoretical approaches to R-loci and identifies the conceptual gaps that motivate a structural reanalysis. Section 3 develops the internal structure of R-loci and introduces Locus-Checking as a general licensing mechanism, specifying its application conditions and limits. Section 4 discusses the theoretical consequences and open questions raised by the proposal, situating it with respect to existing approaches to agreement, null arguments, and discourse reference in sign languages.

2. R-loci in descriptive and theoretical work

Research on sign languages has long recognized that spatial loci play a central role in reference tracking and argument interpretation. There is broad agreement on this point. What remains far less settled is how the grammatical nature of R-loci should be understood. Across the literature, R-loci have been approached from different theoretical perspectives, which differ not only in their empirical focus, but also in the level of grammatical abstraction at which loci are assumed to operate. As a result, there is no unified account of what an R-locus is as a grammatical object or how it participates in argument licensing.

At a general level, four recurring conceptual tendencies can be identified. R-loci have been analyzed as spatial positions in signing space, as abstract indices mediating coreference, as elements involved in verbal agreement, and as discourse-related or gestural devices regulating accessibility. These perspectives are not mutually exclusive. Each isolates a genuine component of how spatial reference functions in sign languages. At the same time, they make different assumptions about where R-loci are represented in the grammar and how their contribution to interpretation is constrained.

R-loci as spatial coordinates

One influential line of research treats R-loci primarily as positions in signing space. These positions are established through pointing or localization and can later be reused for anaphoric reference. This view originates in early descriptive work and is closely associated with analyses that emphasize the embodied and spatial nature of reference in sign languages (Lillo-Martin & Klima 1990; Liddell 1990, 2003). Within this tradition, R-loci function as spatial anchors that support discourse cohesion. Referential expressions introduce entities by associating them with locations in signing space, which can then be accessed by pointing, eye gaze, or spatial modification of the verb. This approach has been particularly successful in capturing gradient and context-sensitive aspects of spatial reference.

The limitation of this perspective lies in its treatment of grammatical structure. While spatial loci are shown to function as anchors for reference, it often remains unclear what kind of grammatical objects they are. They may be treated as syntactic objects, as discourse representations, or as elements of a gestural system that interfaces with grammar. As a result, the relation between spatial localization and argument structure is not formally constrained. In particular, no explicit mechanism is provided by which spatial loci license verbal arguments or interact with verbal valency. What this line of work establishes, however, is that spatial localization is systematic and grammar-relevant, not accidental or purely pragmatic.

R-loci as abstract indices

A second perspective abstracts away from spatial realization and treats R-loci as abstract indices or variables. On this view, spatial loci are understood as the modality-specific realization of underlying indexical information, comparable to indices used in the analysis of anaphora in spoken languages (Meier 1990; Sandler & Lillo-Martin 2006). This approach captures the formal role of R-loci in mediating coreference relations and allows sign language reference to be integrated into general theories of binding and interpretation. Spatial realization is treated as secondary, while the core grammatical work is attributed to index matching.

This abstraction constitutes a fundamental insight that the present proposal fully adopts. Treating spatial loci as instantiations of indexical identity correctly captures their role in establishing formally constrained dependency relations and explains why similar interpretive patterns arise across modalities. In this respect, index-based approaches succeed in isolating a grammatically relevant dimension of reference that is independent of its spatial realization.

At the same time, this perspective also leaves open how indices are represented in the grammar. Indices are typically assumed to annotate expressions rather than to constitute grammatical objects in their own right. As a consequence, the internal structure of R-loci remains implicit, and their role in argument licensing is not formally specified. What this line of work nevertheless makes clear is that indexical identity is an indispensable component of the grammatical architecture underlying R-loci.

R-loci and agreement

A third strand of research situates R-loci within the domain of verbal agreement. Directional or agreement verbs are analyzed as encoding agreement with their arguments by moving between spatial loci associated with subject and object positions (Padden 1988; Engberg-Pedersen 1993; Pfau et al. 2018). On this view, R-loci are tightly integrated into clause structure. Loci associated with arguments determine the form of the verb, and verbal movement is commonly assumed to reflect morphosyntactic agreement. This approach captures robust generalizations concerning verb classes and argument realization.

At the same time, agreement-based analyses face well-known difficulties. Not all verbs that show argument-dependent interpretation exhibit agreement morphology. Moreover, many interpretive effects associated with agreement verbs also arise in the absence of overt spatial modification. The feature inventory involved in agreement remains controversial, and it is often unclear whether R-loci should be treated as agreement features, as agreement controllers, or as the morphological reflex of a more abstract dependency relation. As a result, agreement-based accounts do not generalize naturally to plain verbs, null arguments, or first-person interpretations with body-anchored verbs.

To address this mismatch, Lourenço and Wilbur (2018) propose that agreement in sign languages should be identified not with verbal path movement, but with the co-localization of a verb's articulation site with an established R-locus. On this view, even plain verbs may exhibit agreement insofar as they are produced at a location associated with a discourse referent. While this proposal significantly broadens the empirical coverage of agreement-based accounts, it also raises the question of whether co-localization should be treated as agreement in the narrow morphosyntactic sense. What the work does show, however, is that R-loci interact systematically with verbal structure and cannot be reduced to discourse-level representations.

R-loci as discourse-related devices

Finally, a number of approaches emphasize the role of R-loci in discourse organization and information structure (Liddell 2003; Cormier et al. 2013). On this view, R-loci regulate accessibility and referential persistence and guide interpretation in interaction with non-manual cues such as eye gaze or body lean. The use of signing space is treated as gradient and context-sensitive rather than as the realization of discrete grammatical features.

This perspective captures important pragmatic generalizations, but it typically refrains from assigning R-loci a clear grammatical status. The boundary between grammatical licensing and discourse inference remains blurred. In particular, such approaches do not explain why certain argument positions can remain unexpressed while others cannot, or why discourse accessibility interacts systematically with verbal valency and verb class. At the same time, they correctly highlight that reference tracking in sign languages cannot be understood without considering discourse dynamics.

Taken together, existing approaches converge on the insight that R-loci are central to reference and argument interpretation in sign languages. They diverge, however, in how the grammatical nature of these entities is conceptualized. Each perspective isolates a genuine component of the phenomenon—spatial grounding, indexical identity, interaction with verbal morphology, and discourse accessibility—but none provides an explicit account of how these components are structurally related. As a result, the mechanisms by which R-loci participate in argument licensing remain implicit. The task, therefore, is not to refine any one of these perspectives in isolation, but to develop a structural model of R-loci that makes their grammatical status explicit and integrates their spatial, indexical, and referential properties within a single account. The following section takes up this task.

3. R-Loci and Locus-Checking

As the preceding section has shown, a concrete model of the internal structure of R-loci is required in order to resolve the open questions and inconsistencies identified in existing analyses. This section addresses this gap by proposing a structural characterization of R-loci. I argue that R-loci are syntactic objects that function as minimal units of discourse reference, mediating between argument structure and interpretation. The proposed model is set up as follows:

R-Locus Architecture

An R-Locus consists of two distinct components:

(1) an ***index slot****, which provides an abstract identifier for an argument position and may be specified by a spatial coordinate;*

(2) a ***reference slot****, which hosts referential information and establishes a semantic link to a discourse referent.*

In this respect, the present model differs from earlier accounts in which R-loci are often treated either as purely discursive markers or as mere spatial coordinates. Crucially, in the reconceptualized sense adopted here, an R-locus always comprises both dimensions, even if they are not fully specified in a given instance.

At this point, it is essential to draw a terminological distinction between index and spatial coordinate. The index itself is an abstract label; in order to be specified, it needs to be "annotated" with a concrete spatial coordinate. However, neither the index nor the spatial coordinate in signing space constitutes the R-locus. Rather, the R-locus *has* an index, which may be spatially specified. Similarly, the R-locus *has* a reference slot that can be filled with referential information. Yet neither the discourse referent itself nor the reference slot is identical to the R-locus. Both are components of a more complex syntactic object.

It is precisely this degree of conceptual separation that allows to resolve the inconsistencies observed in the existing literature in a principled way. Once these distinctions are made explicit, it becomes clear that an R-locus, as the minimal unit of reference, always needs both dimensions. And it is exactly the variation in the specification of these two dimensions that gives rise to the observable differences in argument licensing.

In order to capture the variation systematically, it is useful to distinguish the possible states of R-loci, classified according to the degree of specification of their two dimensions.

Specification States

Indexically specified R-Locus:
The index of the R-locus is specified by a spatial coordinate, while the referential component remains unspecified. Instances of this type in DGS include INDEX, PAM, as well as some R-loci encoded in verbal articulation.

Referentially specified R-Locus:
The referential component is specified, while the index remains unspecified. This configuration is assumed to be the basic structure of all NPs. Physically present referents are likewise initially only referentially specified.

Fully specified R-Locus:
The index is specified by a spatial coordinate and linked to a specified referent, yielding a complete and 'active' R-locus that can function as an anchored discourse referent.

Null R-Locus:
The R-locus is neither indexically nor referentially specified. While structurally well-formed, such a configuration cannot function as a free discourse anchor and only becomes interpretable in structural interaction with other R-loci, for example when embedded in a larger configuration such as verbal argument structure.

It is important to clarify that these do not in fact constitute different *classes* of R-loci. In principle, any R-locus can interact with any other. The conditions governing such interactions are, however, non-trivial and will be addressed in the following subsection. It is useful to note, though, that R-loci may enter the derivation in different ways. In particular, a distinction will be made throughout this paper between R-loci that are projected by the verb and R-loci that are introduced independently into the discourse.

The motivation for treating R-loci as inherently two-dimensional does not stem from the presence of fully specified discourse anchors, but from the need to represent structurally present positions that lack both indexical and referential specification. Empirically, such positions are required to account for configurations traditionally described as null arguments: argument positions that are grammatically projected and interpreted, despite the absence of overt indexical marking and independent reference.

In order to represent such configurations, the grammar must make available a minimal structural object that already integrates index and reference as internal dimensions, even when neither is specified. The notion of a **null R-locus** is introduced to capture this extreme configuration.

A null R-locus thus represents the minimal structural instantiation of an argument position, not a reduced or implicit pronominal expression. Rather, it makes explicit that index and reference cannot be treated as independent building blocks that are only combined once both are present. At the same time, a null R-locus cannot function as a free discourse anchor. Precisely because it lacks independent specification along both dimensions, it is structurally dependent and must be embedded in a larger configuration to become interpretable, paradigmatically in structural connection to a verb.

What should also be emphasized at this point is that the simple idea of "binding a reference to an index" does not capture the proposed R-locus model with sufficient precision. Rather, the connection between an indexical and a referential component is postulated as a fundamental property of R-loci themselves. Index and reference are thus inherently linked from the start. What varies is merely the degree of specification of the two components. Only when both components are specified—when the index is associated with a spatial coordinate and the reference slot is filled with concrete referential information—does a fully specified R-locus emerge.

Furthermore, it must be stressed once again that physically present referents do not automatically provide a fully specified R-locus. This point is crucial for a correct interpretation of verbal morphology and for analyses of first-person reference and the phenomenon commonly referred to as *body as subject*.

With the structural conception of R-loci outlined here, it becomes possible to capture both the discourse units attested in DGS and the principles governing argument licensing within a single framework. The mechanism responsible for this interaction is introduced as follows.

Locus-Checking as a general licensing principle

The different specification states of R-loci introduced in the previous section show that argument positions in DGS may enter the derivation with varying degrees of structural completeness. However, structural existence alone is not sufficient for interpretation. For an argument position to be interpretable, missing information must be supplied in a systematic way. The present proposal assumes that this requirement is met by a single, uniform mechanism: the structural *completion* of R-loci.

The process by which an R-locus becomes fully specified will be referred to as **Locus-Checking**. The term 'checking' is not intended to denote mere verification, but rather a systematic operation that fills unspecified dimensions of an R-locus. Locus-Checking can therefore be defined as follows:

> **Locus-Checking:**
> *A structural operation by which two R-loci align their index and reference information in order to fill unspecified slots.*

In this regard, Locus-Checking is conceived as a syntactic operation applying during the derivation, rather than as a post-syntactic interpretive process. Unlike feature-checking, Locus-Checking does not require full identity or feature matching. The only relevant condition is that unspecified information can be supplied through interaction with other R-loci supplying the respective specifications. Accordingly, the basic precondition for Locus-Checking can be stated as follows:

Precondition for Locus-Checking:
At least one of the two interacting R-loci contains an unspecified slot.

This condition does not require both R-loci to be unspecified. As long as at least one R-locus provides an open slot, Locus-Checking may apply. The requirement is thus genuinely minimal. Throughout this paper, a distinction is drawn between *unspecified* and *underspecified* slots. Slots are unspecified at the point of projection whenever no information has yet been assigned to the respective slot. Importantly, this notion is slot-relative: verbal R-loci are projected with an unspecified referential slot, while nominal R-loci may be projected with an unspecified index slot. During Locus-Checking, slots may become underspecified, in the sense that partial information is available while final resolution is deferred until later stages of the derivation.

The basic principle of Locus-Checking can be illustrated by the spatial establishment of discourse referents. In such cases, an NP and an INDEX sign jointly give rise to a single, fully specified R-locus that functions as an anchored and discourse-persistent referent. Structurally, this effect follows from Locus-Checking between two partially specified R-loci. The NP introduces an R-locus whose referential slot is specified, while its index slot remains open. Conversely, the INDEX sign contributes an R-locus with a specified index but no referential information. When INDEX is executed, Locus-Checking aligns the two R-loci, allowing their complementary slots to be filled. The result is a fully specified R-locus that remains available for subsequent reference.

From a formal perspective, successful Locus-Checking could be taken to result in two fully specified R-loci. However, because these loci coincide in all indexical and referential properties, they are indistinguishable for further discourse purposes. Accordingly, the present model assumes that successful checking yields a single active R-locus rather than two independent objects.

It has to be pointed out, though, that this case already represents a special configuration of Locus-Checking. The fact that two R-loci may mutually specify one another is not a prerequisite for the operation. What is crucial is merely that at least one previously unspecified slot can be filled. Beyond spatial establishment as a means of discourse introduction, Locus-Checking is therefore equally relevant for the licensing of verbal arguments. At this point, a uniform structural assumption can be made for all verbs:

Verb Valency and projected R-Loci:
Verbal valency determines the number of R-loci projected by a verb; these loci are referentially unspecified but may differ in index specification across verbs.

This notion has far-reaching consequences for how verbal arguments are conceived. In particular, the presence of a verbal R-locus is not tied to its morphophonological visibility in verbal articulation. Nor is there any requirement that all verb-related R-loci share the same properties. On the contrary, it can be shown that this is typically not the case. The lexically determined specification state of the index component may vary across verbs—and even across different R-loci associated with the same verb. What all verbal arguments have in common, though, is the basic property that they are represented by referentially unspecified R-loci. In this sense, the projection of an argument position introduces a structural dependency on a referential anchor that must be resolved in the course of the derivation. On this basis, the core mechanism of argument licensing can be formulated as a specific instance of Locus-Checking:

Argument Licensing:
An argument is licensed if its referential slot can be filled through successful Locus-Checking.

Throughout this section, the term ‘argument licensing’ is retained as a convenient descriptive label. Importantly, it does not refer to an independent grammatical operation or to a predefined class of arguments, but to the structural completion of verb-associated R-loci via Locus-Checking. The general principle can thus be stated as follows. The articulation of a verbal predicate triggers Locus-Checking for all of its associated R-loci. If these R-loci can be referentially specified, all arguments count as licensed, and the utterance is grammatical. It is important to point out that argument licensing in this model does not follow directly from interpretive recoverability or discourse salience, but from the structural completeness of projected R-loci. Crucially, as R-loci projected by the verb are introduced with an open referential slot, they can never be referentially specified within the verbal projection itself. Referential content can only be supplied by R-loci that are introduced independently of the verb and already carry referential specification. In this sense, independently introduced R-loci constitute the sole source of referential information available for argument licensing.

This uniform characterization makes it possible to capture the structure and functioning of all verb types in DGS within a single framework. Differences arise only with respect to the phonetic realization of verbs and, consequently, the degree to which the index slots of their R-loci are pre-specified. At the surface, this gives rise to what appear to be different “verb classes”.

The number of R-loci associated with a verb is determined by semantic valency: the verb’s lexical specification determines how many R-loci are projected and, accordingly, how many argument positions must be licensed. These verbal R-loci are always unspecified with respect to reference. Through the concrete execution of the verb—determined lexically and semantically—their index values may vary within the range allowed by the verb’s lexical specification.

For example, the index of a body-anchored verb is necessarily specified with respect to a spatial coordinate, namely the position of the signing body. This specification alone, however, does not ensure either licensing or reference. Rather, the execution of the verb triggers Locus-Checking. If the verbal R-locus can only be matched with the referential information associated with the signer, the resulting interpretation corresponds to first person reference.

Standard configurations of Locus-Checking

To make the interactions that were laid out more explicit, the following section summarizes the standard configurations under which Locus-Checking applies:

Spatial establishment (INDEX)
A referentially specified R-locus—introduced by an NP or by a physically present entity—whose index slot is unspecified can be matched with the R-locus contributed by the INDEX sign. The INDEX sign provides a specified index but does not carry referential information. Since both R-loci contain open slots, Locus-Checking allows the missing information in each to be supplied by the respective other R-locus. The result is a fully specified, active R-locus, which is thereby established as a new discourse referent.

Agreement verbs
Agreement verbs assign indexical loci to two of their argument positions, that is, they project R-loci whose index slots are specified while their reference slots remain empty. The execution of the verb triggers Locus-Checking, which evaluates whether the indices of the verbal R-loci coincide with those of already established, fully specified R-loci in the discourse. If this condition is met, the referential information of the established R-loci can be transferred to the open reference slots of the verbal R-loci. As a result, these R-loci become referentially specified, rendering the corresponding arguments licensed and interpretable.

Body-anchored verbs

For body-anchored verbs, Locus-Checking primarily targets the nearest available fully specified R-locus. Since the articulation site of the verb necessarily coincides with the spatial location associated with the signer, the signer's body provides a default source of referential completion. In the absence of additional referential information, this R-locus is therefore automatically inserted into the open reference slot of the verbal R-locus. The outcome is the empirically well-known first-person interpretation.

Plain verbs

The argument positions of plain verbs are projected exclusively as null R-loci. Consequently, they do not require any matching of index information. If a suitably specified referential R-locus is already available in the discourse, it may complete the verbal R-locus via Locus-Checking. If, however, a different referent is intended, or if multiple referential R-loci are simultaneously available in the discourse, additional strategies are required, such as the use of INDEX or PAM.

Taken together, the preceding overview has illustrated how variation in realization and interpretation correlates with differences in the configuration and specification of R-loci. Rather than treating these patterns as properties of distinctive verb classes, the present analysis relates them to more general structural dependencies. With this descriptive groundwork in place, the discussion can now move beyond verb-specific patterns and turn to the basic assumptions and principles that underlie Locus-Checking and its role in structuring argument-related dependencies.

It should be emphasized that spatial establishment is not restricted to non-present referents. Pointing to a physically present individual does not amount to mere deixis or the retrieval of an already given locus. Within the present model, such pointing equally gives rise to the creation of a new, fully specified discourse R-locus. The reason is that physical presence alone provides referential information but does not supply indexical specification. As a result, even physically present referents must undergo spatial establishment in order to function as structurally accessible anchors for Locus-Checking.

The spatial establishment configuration furthermore reveals a structural distinction that cuts across argument-related configurations. Indexical R-loci introduced by INDEX differ in a crucial respect from indexical R-loci contributed by verbs. In both cases, Locus-Checking involves the interaction of an indexically specified R-locus with a referentially specified one. However, verbal indexical R-loci are projected as part of the verb's argument structure and serve exclusively to license verbal arguments.

By contrast, INDEX introduces an indexical R-locus that is not tied to any argument position. When such a locus undergoes Locus-Checking, the result is a fully specified and discourse-persistent R-locus, which remains available for subsequent reference independently of the predicate that triggered its establishment.

This configuration exhibits properties commonly associated with topics. Fully specified and discourse-persistent R-loci display increased accessibility and stability across subsequent operations and thus function as privileged anchors for reference. Rather than encoding topicality by means of a dedicated syntactic feature or projection, the present analysis treats these effects as a consequence of structural completeness and persistence. Topicality, in this sense, is not a primitive category, but an emergent property of R-loci that are independently introduced and fully specified.

The distinction between verb-associated and independently introduced R-loci thus reflects a difference in derivational status and licensing potential, rather than a categorical distinction between verbal and discourse locus types and crucially does not presuppose that the grammar identifies or marks certain R-loci as topics. Independently introduced R-loci are not a separate representational class, but simply R-loci that are available as carriers of referential information beyond the local domain of the predicate. When terms such as *discursive R-locus* or *referential R-locus* are used in what follows, they are intended purely descriptively and do not imply a distinct grammatical category.

Building on these assumptions, the question is no longer what kinds of R-loci are involved in argument licensing, but how relations between R-loci are established in the course of a derivation. While the distinction between verb-associated and independently introduced R-loci captures an important asymmetry in derivational status and licensing potential, it does not by itself determine how an argumental R-locus is completed in concrete configurations. In particular, the model must specify how a suitable referential R-locus is identified for Locus-Checking when multiple candidates are simultaneously available in the discourse.

In a restricted set of configurations, however, the relation required for Locus-Checking is independently constrained. This is the case whenever two R-loci share indexical specification, that is, when the articulation site of an agreement verb coincides with the spatial location of an independently established discourse referent. Under these conditions, Locus-Checking is guided by identity of index values, yielding a highly constrained and stable matching relation. In this respect, such configurations can be understood as the modality-specific analogue of agreement.

At the same time, configurations involving shared indexical specification represent only a proper subset of the attested argument-licensing patterns in DGS. In many cases, no such shared specification is available. In these configurations, the model must therefore provide additional principles that regulate which R-loci are eligible for Locus-Checking.

The preference relations just outlined become particularly salient in configurations in which indexical and referential information are not contributed by the same source. A central case of this type is provided by body-anchored verbs. Those verbs differ from other verb classes in that their articulation at the signer's body makes indexical information available by default. Crucially, however, this articulation site does not impose a fixed indexical specification. Rather, the signer's body provides a weak default source for anchoring that remains open to further specification in the derivation. As a result, the argumental R-loci projected by body-anchored verbs are not fully specified by the verb itself. While a default anchoring is available, referential completion is achieved through Locus-Checking and may align the argument with an independently established R-locus. Body-anchored verbs thus instantiate configurations in which indexical anchoring is weak and defeasible, and in which preference relations determine whether default values are maintained or overridden.

Mechanisms of R-locus selection

With Locus-Checking in place, the remaining question concerns the selection of R-loci in a given configuration. A comparatively simple approach can be adopted: The selection of R-loci for Locus-Checking follows a minimality principle that takes into account both spatial and temporal dimensions.

Selection proceeds in two steps. Preference is given to those R-loci whose index coordinates in signing space most closely coincide. Where such spatial correspondence is not available, a temporal principle applies: the most recently established R-locus in the discourse is treated as the most active and is therefore preferentially selected for Locus-Checking.

Importantly, the minimality considerations governing Locus-Checking cannot be restricted to candidates that are temporally prior in the linear order of the utterance. There is no principled reason to assume that a suitable R-locus must already have been established at the point where an argumental R-locus is projected. On the contrary, empirical patterns in DGS suggest that indexical specification frequently follows the nominal expression it relates to. INDEX is typically produced after an NP, rather than preceding it, and thus supplies indexical information retrospectively.

This observation has direct consequences for the interpretation of minimality. While spatial proximity remains a primary criterion for accessibility, temporal proximity cannot be understood as strictly directional. Instead, temporal relations are evaluated symmetrically: indexical specification may be supplied either by previously established R-loci or by material that appears later in the discourse, provided that the relevant R-locus is sufficiently accessible at the point of evaluation.

This selection mechanism becomes particularly relevant in configurations that lack any indexical guidance. In such cases, verbal R-loci must be completed without relying on shared specification, and the grammar must therefore select an appropriate referential R-locus solely on the basis of accessibility. Plain verbs provide a particularly clear illustration of this situation, as they project their argument positions without contributing any indexical specification of their own.

As discussed earlier, plain verbs project their argument positions exclusively as null R-loci. In line with all verbal R-loci, these positions require referential completion through Locus-Checking with an independently introduced R-locus in the discourse. As a result, while referential completion is obligatory in all cases, the selection of a suitable referential anchor is less tightly constrained in the absence of indexical guidance. Plain verbs therefore provide a particularly instructive case for examining how the selection mechanism operates when no indexical specification narrows the space of possible matches.

Configurations of this type give rise to interpretive effects that appear comparatively unconstrained at the surface. Since plain verbs do not restrict the selection of a referential anchor by indexical means, different discourse referents may in principle be compatible with a given verbal R-locus. As a result, the dependency between a null argument position and its referential interpretation is not immediately recoverable from overt form alone. This apparent flexibility has motivated the characterization of such configurations as involving "unselective binding" in the literature.

Within the present model, however, this apparent flexibility does not reflect an absence of grammatical constraints. Rather, the relation established through Locus-Checking is regulated by the same selection principles introduced above. Matching is constrained by the availability and accessibility of suitable referential R-loci in the derivation and is guided by spatial and temporal proximity. The dependency is therefore not unselective, but systematically restricted, even in the absence of overt indexical guidance. In sign languages, this dependency is particularly transparent, since discourse referents are spatially anchored and thus more overtly represented.

The Generic R-Locus

The selection principles discussed so far presuppose the availability of at least one referential R-locus that can complete a verbal R-locus via Locus-Checking. However, discourse structure does not always provide such a candidate. Since all verbal R-loci nevertheless require referential completion in order to be licensed, the grammar must make reference available independently of discourse establishment. It will therefore be argued that this requirement is met by a **Generic R-Locus**, which constitutes a reliable structural option within the system rather than a repair strategy.

The Generic R-Locus is a regular element that can participate in Locus-Checking in the same way as discourse-established R-loci. Like all R-loci, it comprises both an index slot and a referential slot. Crucially, neither of these slots is left unspecified; instead, both are positively specified as 'generic'. The Generic R-Locus thus does not represent an unresolved or underspecified argument position, but a fully specified referential anchor with generic content.

The Generic R-Locus is always available in the derivation and does not need to be introduced or established in the discourse. It enters Locus-Checking under the same conditions as any other R-locus and is subject to the same constraints on accessibility and selection.

As a consequence, generic specification is not treated as a last-resort strategy. The selection of the Generic R-Locus over a discourse-established R-locus is governed by preference and competition, not by necessity. This perspective allows generic interpretations to arise both in the absence of suitable discourse referents and in contexts where discourse referents are available but intentionally not selected.

As is already implied by the structural description of the Generic R-Locus, it may not only be recruited to fill the referential slot of a verbal R-locus, but also its index slot. This has important consequences for the interpretation of bare nominals. In configurations involving a verbal R-locus whose index is not independently specified, a simple NP supplies the referential content required for Locus-Checking, while indexical specification can be contributed by the Generic R-Locus. The result is a configuration in which reference is fixed, but the index is specified as generic, yielding an interpretation characteristic of bare nominals. This configuration illustrates more generally that indexical and referential specification need not be contributed by the same source within a given R-locus configuration.

Preference relations and default resolution

Whenever multiple R-loci are eligible for Locus-Checking, the grammar must impose an ordering among competing candidates. Within the present model, this ordering is sensitive to the degree of specification and stability of the available R-loci. Those R-loci in which both indexical and referential information are explicitly specified constitute the strongest candidates for selection.

By contrast, R-loci that provide only partial specification—either by contributing referential information without an independently specified index, or by contributing indexical information without a fully determined referential value—constitute weaker competitors. Such R-loci may enter into Locus-Checking, but they do so under reduced preference.

As a consequence, when a fully specified and discourse-established R-locus is available, it is systematically preferred over less fully specified alternatives, including generic specification. This preference relation does not define a categorical exclusion but gives rise to gradient resolution patterns in which weaker candidates may still be selected under appropriate discourse conditions. The outcome of Locus-Checking thus reflects competition among R-loci that differ in their degree of specification rather than the application of discrete licensing rules.

The preference relations just outlined become particularly salient in configurations in which indexical and referential information are not contributed in a fully aligned way. A central case of this type is provided by body-anchored verbs. In these configurations, the verb makes indexical anchoring available by virtue of its articulation site, without thereby fixing a fully specified index or determining reference. Referential completion must still be achieved through Locus-Checking. Body-anchored verbs therefore instantiate configurations in which default anchoring and referential completion are dissociated, providing a revealing testing ground for how preference and competition among R-loci are resolved. These preference relations are not stipulative but follow directly from the degree of structural specification encoded in the R-locus architecture.

Body-anchored verbs differ from other verb classes in that they are articulated at the signer's body and may therefore make indexical information available by default. Crucially, however, this articulation site does not impose a fixed indexical specification. Rather, both the signer's body and the locus of articulation function as last-resort default values for indexical and referential slots, respectively, which remain open to further specification in the course of the derivation.

While a default anchoring is available in the absence of competing candidates, referential completion is still achieved through Locus-Checking and may align the argument with an independently established discourse R-locus. In the absence of a competing fully specified discourse R-locus, the default values may be used for specification. Under such conditions, the signer's body constitutes the most immediately available anchor for referential completion, yielding interpretations that are commonly described as first-person–like. Importantly, this outcome does not follow from the encoding of a grammatical person feature, but from the general preference relations governing the selection of R-loci.

Where an independently established discourse R-locus is available, however, its greater degree of specification and stability renders it a stronger competitor. In such cases, the default anchoring provided by the body may be overridden, and Locus-Checking aligns the argumental R-locus with the discourse referent instead. The interpretation associated with body-anchored verbs therefore reflects a gradient resolution of competing candidates rather than a categorical person-based constraint.

A different outcome arises when the body-based anchor is maintained as a default source of reference, while indexical specification is supplied by the Generic R-Locus. In such cases, the configuration does not involve the selection of speaker reference but yields a reading in which reference is anchored to a definite human referent without being discourse-identified. Interpretations commonly paraphrased as *somebody* thus follow from the interaction of default referential anchoring and generic index specification and should be kept distinct from deliberate signer reference.

Importantly, however, these distinct configurations are not accompanied by systematic phonological differences. The maintenance of body-based anchoring with generic index specification is surface-identical to configurations in which the body-based default is exploited for deliberate signer reference, as well as to cases in which indexical specification has not yet been resolved. As a result, the grammatical system may temporarily leave the precise interpretive outcome open, allowing subsequent material in the discourse to further constrain the choice among competing R-loci. The following section takes up this residual indeterminacy and examines how interpretive commitments can be refined in the course of a derivation.

Delayed specification and interpretive ambiguity

The configurations discussed so far already point toward a more general consequence of index underspecification. Considering the possibility of temporarily underspecified indices, the model provides a structural explanation for *indefiniteness*. While a definite interpretation is only possible once the index of an R-locus is specified, an indefinite interpretation results from an index that is (not yet) specified. Crucially, *indefinite* must be clearly distinguished from *non-definite*, and in particular from *generic*. An underspecified index is indefinite simply because its specification status is unresolved. A definite interpretation is therefore not excluded in principle, since it may be established later in the discourse through subsequent index specification.

Likewise, an index that appears to be phonologically absent may in fact be specified as 'generic'. In cases of uncertainty, this distinction can only be determined by observing whether a suitable topical R-locus appears later in the discourse that is sufficiently close to permit successful checking. During this phase of the discourse, the index of the argument can be conceptualized as 'floating'.

The notion of a *floating index* also provides a plausible explanation for the well-known difficulties in analyzing body-anchored verbs. If the body-based R-locus competes only with a plain NP in the discourse, neither carries a specified index. When the body-anchored verb is articulated and Locus-Checking is initiated, it cannot be decided immediately which referent is the stronger candidate. A strict application of spatial and temporal minimality would predict that the signer's body should always be preferred in such cases. Empirically, however, this is not consistently observed.

This pattern can be explained by assuming that the competing NP carries an index that is not yet fixed at the point where Locus-Checking is initiated. As long as indexical specification remains unresolved, the referential interpretation of the argumental R-locus is structurally underspecified. In such configurations, no immediate commitment to a particular referential anchor is enforced, since the availability of a more fully specified R-locus cannot yet be excluded.

If subsequent material in the derivation supplies indexical specification—for instance through the use of INDEX—the resulting fully specified R-locus constitutes a stronger candidate under the general preference conditions governing Locus-Checking and may override the weaker body-based default. If no such specification occurs, the default locus remains the only available candidate and is therefore selected. This does not contradict the assumption that unspecified R-loci cannot represent licensed arguments. In the present configuration, specification is guaranteed in any case; it is merely temporarily undecided. These considerations underscore once more that temporal dynamics and transient uncertainty should be treated as natural aspects of linguistic interpretation.

In fact, the dynamics and uncertainty observed so far extend to another domain that is often assumed to be more clear-cut. Within the present model, the traditional contrast between overtly realized arguments, discourse-mediated or null arguments cannot be maintained as a categorical grammatical distinction. Rather than employing a tripartite classification of argument types, the present model replaces this contrast with a gradual notion of accessibility based on the distance between the verb and the referential R-locus targeted for checking. Morphophonological marking may influence this accessibility by shaping the search space in which potential referential loci are evaluated, but it does not introduce a separate licensing mechanism. While this reanalysis departs from a familiar classificatory scheme, it does not multiply mechanisms or weaken grammatical constraints. On the contrary, it allows patterns of argument realization that are otherwise treated as heterogeneous to be captured within a single, structurally grounded framework.

Structural exclusion conditions

Given the simplicity and generality of Locus-Checking, one might wonder whether the mechanism is sufficiently constrained to function as a regulated grammatical principle. The apparent degrees of freedom may invite the impression of arbitrariness. For this reason, the following line-up outlines several configurations in which successful Locus-Checking is clearly excluded, thereby delineating the limits of the mechanism:

Two fully specified R-loci
Both R-loci already carry a specified index and a specified reference. In this configuration, no Locus-Checking can apply, since there is no dimension that could be filled.

Two null R-loci
Both R-loci contain only unspecified slots. While Locus-Checking may in principle be initiated, it cannot succeed, as no information is available that could complete either locus.

Incompatible specifications
Two partially specified R-loci are incompatible if they are specified with different values in the same dimension. For example, if both R-loci carry a filled index with different spatial coordinates, successful Locus-Checking is excluded.

Locus-Checking is therefore not arbitrary. It follows clear regularities, only one aspect of which resembles agreement in spoken languages. Precisely because Locus-Checking operates in a much more general way, it is suitable as an overarching principle of argument licensing, applying both to verbal morphology and to the analysis of null arguments. Crucially, Locus-Checking does not need to be actively “licensed” by a particular feature configuration. It is sufficient that it is *not blocked*. The possibility of successful Locus-Checking thus cannot be characterized positively, but only by the absence of exclusion criteria.

The catalogue of exclusion configurations presented here does not claim to be exhaustive. Rather, it should be understood as a starting point for further theoretical development. It is likely that additional, more fine-grained exclusion criteria will be required to capture other observable phenomena more adequately. The proposed model also entails a further theoretical and terminological consequence that should be made explicit at this point:

INDEX is not an index. Given the structure of R-loci and the terminology adopted here, the INDEX sign itself cannot be identified with an index in the sense of the R-locus structure. Nevertheless, the function of INDEX remains closely tied to indexical information. Since INDEX contributes an indexical R-locus without specified reference, its central ability to "link referents to locations" follows directly. Structurally, the same holds for PAM, which may even provide indexical information for two argument positions.

It is therefore proposed to provisionally group these elements under the label **index markers**. This terminology remains close to the traditional notion of indices, while allowing for an intuitive integration into the present model. Potentially, non-manual markers such as eye gaze or body lean may also belong to this class of index markers. Determining this, however, requires further empirical investigation and cannot be settled conclusively at the current stage of the model. For present purposes, it suffices to note that such elements contribute indexical specification without introducing independent referential content.

Against this background, the conditions governing successful Locus-Checking can now be made explicit. Argument licensing in the present model is tied to the referential completion of verb-associated R-loci. What is decisive is that the relevant reference slot is filled through Locus-Checking—either by linkage to an independently introduced, discourse-bound R-locus or by recruitment of a generic operator. The index component may remain underspecified, yielding an indefinite interpretation, or may itself be generically specified; it may also be specified retroactively in the course of the discourse. Among other things, the analysis shows that configurations traditionally described as null arguments are not exceptional, but regular outcomes of the same structural mechanism, despite potentially differing in their underlying structural configuration

At the same time, the present analysis deliberately abstracts away from several dimensions that call for further refinement. Open questions include the precise spatial and temporal domains within which Locus-Checking can apply, the conditions under which floating indices are resolved, and the systematic integration of thematic relations. These issues are not treated as residual problems, but as points of principled extension that require further theoretical and empirical work beyond the scope of this paper.

The contribution of the present proposal is primarily architectural. Its empirical content lies in the restrictions it places on the space of possible R-locus configurations, rather than in the formulation of construction-specific predictions. As a result, many of its empirical consequences remain implicit and become fully testable only once further dimensions of the model—such as thematic differentiation—are integrated.

What remains to be addressed concerns not the internal functioning of the model, but its scope, its empirical grounding, and its relation to existing approaches. The following section takes up these questions.

4. Empirical motivation and theoretical implications

The analysis developed in the preceding sections provides a structural basis for relating variation in argument realization and interpretation to the configuration of R-loci. Once argument positions are treated as verb-associated R-loci that require referential completion, interpretive differences no longer need to be attributed to distinct argument types or construction-specific mechanisms. Instead, they follow from the structural accessibility of referential R-loci and from the conditions under which Locus-Checking applies.

From this perspective, differences in interpretation reflect the interaction between verbal structure and independently introduced referential R-loci, rather than the presence or absence of particular features or categories. The following discussion illustrates how this structural approach bears on a range of phenomena that have figured prominently in the literature.

A central analytical advantage of the model lies in its ability to distinguish systematically between different sources of interpretive variation. Rather than appealing to separate feature inventories or argument types, interpretive differences follow from the structurally regulated accessibility of referential R-loci. Verb-associated R-loci require referential completion, and whether a suitable referential R-locus can be recruited depends both on its internal specification and on its structural distance from the predicate. As a result, phenomena that are often described as optional argument realization or discourse-driven interpretation follow directly from how referential R-loci are structurally related to the verb.

This perspective proves especially useful for the analysis of body-anchored verbs. Rather than stipulating a privileged grammatical status for first person or treating body reference as inherently deictic, the model derives first-person interpretations from the interaction of partially specified R-loci and general selection principles. At the same time, it predicts the observed variability in the presence of overt nominal competitors and explains why fully established spatial loci carry the potential to systematically override body-based reference, while simple NPs do not.

Similarly, the analysis of plain verbs and null arguments benefits from a structural reinterpretation. Instead of assuming invisible pronominal elements or deletion processes, null arguments are treated as regular instances of underspecified verb-associated R-loci whose referential slots are completed through Locus-Checking, either by successful checking with an independently introduced referential R-locus or with a Generic R-Locus. This allows generic and discourse-dependent readings to be handled within the same licensing framework and avoids the need to posit language-specific null categories. At the same time, the selection principles invoked in this analysis are not intended to yield a fully deterministic resolution procedure. Rather, they delimit the space of structurally eligible R-loci, within which additional factors may further constrain interpretation in concrete discourse configurations.

Against this background, the following remarks briefly situate the proposal with respect to the data on which it is based. The empirical foundation of the present proposal consists in a qualitative investigation of argument omission and referential interpretation in German Sign Language (DGS) developed in my Master's thesis (Nazar 2025), which serves as the point of departure for the theoretical reconstruction advanced here. While this empirical basis is necessarily limited in scope, the model has proven capable of capturing a range of well-attested patterns in a unified way and of making explicit predictions about the interaction of verbal structure, spatial reference, and interpretation.

In this sense, the primary empirical contribution of the model lies not in exhaustive coverage, but in the structural coherence it brings to a set of phenomena that have previously been treated in a largely fragmented manner.

Relation to existing approaches: null arguments as a diagnostic domain

In order to situate the present proposal with respect to existing approaches, null arguments provide a particularly revealing area of investigation. In this regard, the R-locus model developed in this paper intersects with several strands of research in sign language linguistics, while differing from them in important respects. Most existing accounts of argument realization in sign languages fall into one of three broad classes: agreement-based analyses, approaches that rely on null pronominal elements, and discourse-pragmatic models that treat argument omission primarily as an interpretive phenomenon. The present proposal does not deny the empirical insights of these approaches, but it reframes their core observations within a different structural perspective.

Agreement-based analyses typically assume that spatial modification of the verb reflects morphosyntactic agreement with its arguments. While such accounts successfully capture systematic correspondences between movement and reference, they face persistent difficulties in defining the feature inventory involved, in accounting for asymmetries between verb classes, and in explaining why similar interpretive effects arise in the absence of overt agreement morphology. By contrast, what appears as agreement is not treated as a primitive relation in the presented model, but a surface manifestation of a specific instance of a more general matching operation.

Analyses that posit null pronominal elements, such as *pro*, aim to preserve standard assumptions about argument structure by allowing syntactically projected arguments to remain phonologically unrealized. While this strategy offers a straightforward way of encoding missing arguments, it leaves open the question of how such elements are licensed and interpreted in the absence of overt agreement marking. In contrast, the R-locus model does not assume invisible pronominal categories. As a result, argument omission does not require additional syntactic objects but follows directly from the general architecture of verbal R-loci and their interaction with referential R-loci. Within this perspective, arguments are not taken to be primitive grammatical objects but emerge as effects of how R-loci are structurally completed and aligned.

Purely discourse-pragmatic approaches emphasize the role of contextual accessibility and salience in the interpretation of missing arguments. These factors are clearly relevant and are not denied by the present analysis. However, treating argument omission as a matter of discourse alone makes it difficult to capture the systematic interaction between verbal form, spatial structure, and interpretive constraints observed in sign languages like DGS. Within the present model, discourse effects can be stated in terms of constrained selection among competing R-loci, rather than as post-syntactic interpretive repair.

Taken together, the proposed R-locus model occupies an intermediate position between morphosyntactic and discourse-based approaches. It preserves the insight that argument interpretation is structurally constrained, while avoiding strong commitments to agreement features or null categories. At the same time, it makes explicit how discourse information enters the derivation, without relegating argument licensing to pragmatic inference alone. In this way, the model provides a unifying perspective on phenomena that have traditionally been treated as theoretically heterogeneous.

Scope and open questions

Several aspects of the model remain open for further refinement. One concerns the selection principles governing Locus-Checking. While spatial and temporal minimality have been argued to play a central role in regulating the interaction of R-loci, the precise delimitation of the relevant domains is left underspecified. In particular, the notion of temporal proximity raises questions about how grammatical structure interacts with memory-related constraints. The concept of a floating index provides a way of modeling temporary underspecification, but the conditions under which such indeterminacy is resolved require further empirical and theoretical investigation.

A related issue concerns the interaction between the proposed licensing mechanism and lexical variation. Although apparent verb class differences are derived from variation in index specification and articulation, the present analysis does not attempt to provide a comprehensive typology of verb classes. Further work is needed to examine how these patterns interact with lexical semantics, constructional restrictions, and cross-signer variation. Similarly, the potential role of non-manual markers as index markers has only been touched upon and remains an open empirical question.

It should also be emphasized that the R-locus model is not intended as a general theory of discourse reference. While discourse accessibility and topicality play a role in the selection of matching R-loci, the analysis is restricted to those aspects of reference that are grammatically encoded in the internal structure of R-loci themselves. Broader pragmatic factors, such as world knowledge or conversational inference, lie outside the scope of the present proposal.

Finally, although the present analysis has been developed on the basis of data from German Sign Language, the structural principles it attributes to R-loci are not assumed to be specific to DGS. Rather, the proposed architecture is intended to capture properties of spatial reference that are characteristic of sign languages more generally. The present paper does not pursue a systematic cross-linguistic comparison. But the separation of index and referential information and the treatment of argument licensing as a form of structural completion suggest a perspective that extends beyond individual sign languages.

At a more abstract level, these considerations raise the further question of whether the structural logic underlying R-loci is tied to the visual–spatial modality at all. For the purposes of such a broader abstraction, the term **R-Anchor** may be used to refer to a modality-independent anchoring structure of which R-loci constitute one possible instantiation, characterized by spatial indices. Exploring this possibility in detail, however, goes beyond the scope of the present study.

In sum, the R-locus model should be understood as a focused contribution to the analysis of reference and structural argument licensing in sign languages. Its primary goal is not to offer a comprehensive theory of argument structure, but to make explicit the structural role of R-loci and to show how a small set of assumptions can bring coherence to a range of previously disparate observations. Seen this way, argument structure does not require an inventory of primitively distinct elements but can be derived from relations among positions that are uniform in their basic architecture. Further empirical work will be required to refine the model, test its predictions, and explore the extent to which the proposed mechanisms generalize beyond the domain examined here.

This seems a worthwhile endeavor.

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